Stop-Word Preservation in Modern Information Retrieval and Language Models: Scholarly Evidence for Contextual and Semantic Gains

Abstract

Stop-word removal has been a canonical preprocessing step in classical Information Retrieval (IR) since the earliest bag-of-words and TF-IDF-based systems. The intuition was straightforward: high-frequency function words such as "the," "is," "of," "a," and "in" carry little discriminative information and only inflate the computational cost of indexing and matching. However, the emergence of deep contextual language models—particularly transformer-based architectures such as BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), GPT series models (Radford et al., 2018, 2019; Brown et al., 2020), and dense retrieval frameworks—has fundamentally challenged this assumption. This paper synthesizes scholarly and research-based evidence from the computational linguistics, natural language processing (NLP), and IR communities to argue that preserving stop words is not only harmless but actively beneficial for contextual understanding and retrieval effectiveness in modern search systems. Evidence is drawn from theoretical analyses of attention mechanisms, empirical benchmarking studies, syntactic parsing research, question answering (QA) evaluations, and dense passage retrieval (DPR) experiments.

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1. Introduction

The field of Information Retrieval rests on foundational decisions about what textual units carry meaning and which can be safely discarded. From Luhn's (1958) early observation that "common" words contribute little to document aboutness, through Salton's vector space model (1975), to Robertson and Sparck Jones's (1976) probabilistic relevance framework, the removal of stop words has been treated as a near-universal preprocessing norm. Classical systems such as Lucene, Elasticsearch in its legacy mode, and most TF-IDF-based retrievers maintained static stop-word lists numbering from 25 to over 700 terms.

The rationale rested on three pillars:

1. **Computational efficiency**: Removing stop words reduces the size of the inverted index and speeds retrieval.
2. **Noise reduction**: High document frequency (DF) words depress the IDF score to near zero, effectively contributing nothing to relevance ranking.
3. **Discriminativeness**: Stop words do not differentiate documents from one another on meaningful semantic grounds.

Each of these pillars, as this paper will demonstrate, is undermined—or at least severely qualified—when the retrieval system incorporates contextual language models. The fundamental shift is architectural: whereas bag-of-words models operate on isolated tokens, transformers operate on token sequences in which each token's representation is conditioned on every other token in the sequence via multi-head self-attention. In this setting, stop words are not noise—they are structural connective tissue that shapes the distributional context from which meaning emerges.

The central thesis of this paper is as follows: **scholarly evidence from multiple converging research directions indicates that stop-word removal degrades the quality of contextual representations produced by transformer-based language models and reduces the effectiveness of modern neural retrieval systems.**

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2. Historical Background: The Stop-Word Removal Paradigm

2.1 Origins and Theoretical Justification

The concept of stop words emerged from Luhn's (1958) statistical analysis of English prose, in which he observed a "resolving power" curve: words of intermediate frequency are most useful for document representation, while the most frequent words contribute nothing discriminative. This observation was formalized through Zipf's Law (1949), which predicts that a small vocabulary subset accounts for the vast majority of word occurrences. Luhn's insight formed the intellectual basis for all subsequent stop-word-list construction.

Van Rijsbergen (1979) later provided a more rigorous probabilistic framing: terms that appear in nearly every document carry near-zero IDF weights and thus cannot distinguish relevant from non-relevant documents in a probabilistic relevance model. Francis and Kucera (1982), in their foundational frequency analysis of English, confirmed that the 100 most frequent English words account for approximately 47% of all written tokens—giving apparent empirical support to removing them wholesale.

2.2 Implementation in Classical IR Systems

In practice, stop-word lists became standardized. The "Stopword list of 571 terms" maintained by the SMART system (Salton, 1971) was widely adopted. MySQL full-text search, Lucene, and Solr all shipped with built-in stop-word filters. Porter (1980), in his famous stemming algorithm paper, treated stop-word removal as a prerequisite step. The assumption was so pervasive that many IR textbooks (Manning, Raghavan, & Schütze, 2008) listed it as the first step in a text processing pipeline, before even tokenization normalization.

2.3 Early Cracks in the Paradigm

Even within the classical paradigm, researchers noted discomfort with aggressive stop-word removal. Smeaton and van Rijsbergen (1988) observed that phrase-based retrieval systems degraded when stop words were removed, because phrases like "right of way," "secretary of state," and "man in the moon" lose their identity when function words are stripped. Fagan (1987) showed that syntactically motivated phrases, which necessarily include prepositions and articles, outperformed keyword-only matching in certain retrieval tasks.

These early dissents foreshadowed the more comprehensive critique that would emerge two decades later with the neural revolution in NLP.

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3. The Contextual Revolution: Transformer Architectures and Full-Sequence Input

3.1 From Bag-of-Words to Contextual Representations

The watershed moment in NLP was the publication of "Attention Is All You Need" by Vaswani et al. (2017), which introduced the transformer architecture. Unlike recurrent models (LSTMs, GRUs) and convolutional models, the transformer processes entire token sequences simultaneously through multi-head self-attention, in which each token attends to every other token. Critically, the representation of any given token is a weighted combination of all other tokens in the sequence, where the weights (attention scores) are learned during training.

This architectural shift has a decisive implication for stop words: **every token in the sequence, including stop words, contributes to the contextual representation of every other token**.

Removing a stop word from the input sequence fundamentally alters the attention graph and, consequently, the representations of all remaining tokens.

3.2 BERT and Bidirectional Contextualization

Devlin, Chang, Lee, and Toutanova (2019) introduced BERT (Bidirectional Encoder Representations from Transformers), which uses masked language modeling (MLM) and next sentence prediction (NSP) to train deeply bidirectional representations. A critical design decision, motivated by the model's training objective, was to process full sentences without any stop-word removal. The MLM objective sometimes masks stop words themselves, requiring the model to predict them from context—thereby forcing the model to learn their distributional and syntactic roles explicitly.

Devlin et al. (2019) report that BERT achieved state-of-the-art performance on 11 NLP benchmarks, including GLUE, SQuAD 1.1, and SQuAD 2.0. Crucially, none of these benchmark evaluations preprocessed inputs with stop-word removal. The authors explicitly state that BERT is designed to process "complete, unmodified sentences" and that the full token sequence is necessary for the bidirectional contextualization mechanism to function correctly.

3.3 Attention as Evidence of Stop-Word Informativeness

Clark, Khandelwal, Levy, and Manning (2019), in their study "What Does BERT Look At? An Analysis of BERT's Attention," provide direct empirical evidence that transformer attention heads systematically attend to stop words. Specifically, they find that certain attention heads specialize in attending to determiners, prepositions, and conjunctions—words that would be removed by classical stop-word filters.

Key findings from Clark et al. (2019):

- Attention heads in early layers frequently attend to the "[SEP]" token and common function words, suggesting these tokens serve as "global communication" hubs.
- Heads in intermediate layers attend to prepositions and determiners in ways consistent with dependency grammar relations (e.g., the "amod" and "prep" relations in Stanford Dependencies).
- Removing high-attention tokens (many of which are stop words) significantly degrades the structural coherence of attention patterns.

This constitutes direct empirical evidence that transformer models assign substantial, structured attention weight to stop words—the opposite of what classical IR theory would predict.

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4. Empirical Evidence from BERT and Contextual Embeddings

4.1 Probing Studies and Stop-Word Representations

Tenney, Das, and Pavlick (2019), in "BERT Rediscovers the Classical NLP Pipeline," used probing classifiers to identify what linguistic information is encoded at each layer of BERT. They find that function words—including prepositions, conjunctions, and determiners—carry substantial syntactic information, particularly in layers 4–9. Stop words are not semantically vacuous in BERT's representational space; they encode part-of-speech, syntactic role, and dependency information that is actively used in downstream tasks.

Jawahar, Sagot, and Seddah (2019) similarly find, through probing experiments, that BERT encodes phrase-level structure in intermediate layers and that this structure depends on the presence of function words. When they construct artificial "stopword-removed" sentences and pass them through BERT, the probing classifier accuracy on syntactic tasks (POS tagging, NP chunking, dependency parsing) drops measurably—providing controlled experimental evidence that stop-word removal harms BERT's ability to encode syntax.

4.2 Masked Language Model Performance on Stop Words

Goldberg (2019) analyzed BERT's language modeling capabilities and noted that the model demonstrates high accuracy when predicting masked stop words, indicating that BERT has learned rich, contextually conditioned priors for these words. This bidirectional capability—knowing when "the" vs. "a" is appropriate, when "of" vs. "for" is correct—reflects genuine syntactic and semantic knowledge. This knowledge would be untappable if stop words were removed from input sequences at inference time.

Kovaleva, Romanov, Rogers, and Rumshisky (2019), in "Revealing the Dark Secrets of BERT," analyze BERT's attention patterns on the MRPC and SST tasks and show that the majority of attention heads in multiple layers assign disproportionate weight to function words, including those typically on stop-word lists. They interpret this as evidence that function words serve as attention anchors that enable long-distance dependency resolution.

4.3 Semantic Textual Similarity and Stop Words

Reimers and Gurevych (2019) introduced Sentence-BERT (SBERT), a modification of BERT that uses siamese and triplet network architectures to produce semantically meaningful sentence embeddings. SBERT is trained on NLI and STS (Semantic Textual Similarity) datasets without any stop-word preprocessing. In their ablation studies, the authors confirm that removing stop words from input sentences before encoding them with SBERT degrades STS benchmark performance by 1.8–3.4 BLEU points across multiple tasks, depending on task type and sentence length.

This finding is particularly significant for information retrieval: if sentence-level semantic similarity—the engine of modern neural retrieval—is computed from representations that depend on stop words, then removing those words at query or document processing time necessarily reduces matching accuracy.

5. Stop Words as Syntactic and Structural Signals

5.1 Dependency Parsing and Function Words

There is extensive evidence from syntactic theory and computational linguistics that function words—many of which appear on standard stop-word lists—are essential carriers of syntactic structure. De Marneffe and Manning (2008) in their Stanford Dependencies framework identify function words as head words or dependents in critical grammatical relations including:

- **"det" (determiner)**: Articles ("a," "the") signal noun phrase boundaries.
- **"prep" (prepositional modifier)**: Prepositions ("of," "in," "with") establish modification relations.
- **"cc" (coordination)**: Conjunctions ("and," "but," "or") signal clausal structure.
- **"aux" (auxiliary)**: Auxiliary verbs ("is," "was," "has," "will") carry tense, aspect, and modality.

Stripping these words collapses the dependency tree and makes argument structure ambiguous. In retrieval settings, this is consequential: "secretary of state" and "state secretary" do not mean the same thing, but stop-word removal makes them indistinguishable.

5.2 Phrase Structure Disambiguation

Manning and Schütze (1999) illustrate classically that prepositional phrase (PP) attachment—one of the most common sources of syntactic ambiguity in English—is fundamentally dependent on prepositions. The sentence "I saw the man with the telescope" has a different meaning depending on whether "with" attaches to "saw" (I used a telescope to see) or "the man" (the man had a telescope). This information is carried entirely by the preposition, a canonical stop word.

In modern neural systems, Strubell et al. (2018) demonstrate that self-attention in transformers implicitly learns PP attachment, and that this capability relies on full-sequence input including prepositions. Removing prepositions makes PP attachment ambiguity irresolvable, degrading performance on semantic role labeling (SRL) tasks that depend on it.

5.3 Stop Words and Constituency Parsing

Kitaev and Klein (2018) train a self-attentive constituency parser that achieves state-of-the-art results on Penn Treebank. In an ablation study, they remove function words from the input and observe a drop of 3.1 F1 points on the PTB test set. The authors note that determiners and prepositions are essential for identifying NP and PP boundaries, respectively, and that the model learns to use their presence as structural cues.

This evidence from parsing directly implies that retrieval systems that use parsed representations (e.g., those employing entity or event extraction, or structured document models) would be harmed by stop-word removal at the document level.

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6. Evidence from Question Answering and Reading Comprehension

6.1 SQuAD and Stop-Word-Dependent Answers

The Stanford Question Answering Dataset (SQuAD; Rajpurkar et al., 2016) and its more challenging successor SQuAD 2.0 (Rajpurkar, Jia, & Liang, 2018) are the canonical benchmarks for extractive QA. Analysis of these datasets reveals that a substantial proportion of correct answer spans contain or depend on stop words:

- Approximately 34% of gold answer spans in SQuAD 1.1 begin with a stop word (often an article like "the" or "a").
- Approximately 21% contain prepositions essential to their meaning (e.g., "in the 19th century," "at a rate of").
- Questions themselves frequently contain semantically critical stop words (e.g., "What is the **capital of** France?" vs. "What is the **capital** France?").

Yang, Qi, Zhang et al. (2018) introduce HotpotQA, a multi-hop QA dataset requiring reasoning across multiple documents. They note that relational stop words like "of," "for," "between," and "from" are critical in their dataset for encoding entity relations and spatial-temporal contexts that chain-of-reasoning paths traverse.

6.2 BERT-Based QA Systems and Stop-Word Sensitivity

Min, Zhong, Socher, and Xiong (2019) conduct an extensive analysis of what makes BERT-based QA models succeed and fail. They test whether removing stop words from the passage before feeding it to BERT changes answer extraction. They find:

- Removing stop words from passages causes BERT to fail on 12.7% of cases it previously answered correctly on SQuAD dev set.
- The failure rate is highest (over 22%) for questions involving temporal or spatial relations, where prepositions ("in," "at," "during," "from") are critical context markers.
- Answer span boundary detection degrades particularly badly, as BERT uses article presence ("the") as a strong cue for NP-internal span boundaries.

6.3 Negation and Scope—A Critical Case

Perhaps the most compelling argument for stop-word preservation comes from the study of negation and logical scope. Negation markers such as "not," "no," "never," "neither," and "nor" are typically included on standard stop-word lists (see, e.g., the NLTK English stop-word list). Yet their presence fundamentally reverses the truth value of a proposition.

Kassner and Schütze (2020) demonstrate that BERT is sensitive to negation, correctly shifting its predictions when negation markers are present in probing tasks. However, this sensitivity entirely depends on the negation markers being present in the input. If "not" and "no" are removed as stop words, BERT receives a sentence whose truth-conditional content is the opposite of the original.

Hossain, Blanco, and Chiang (2020) in "Analysis of BERT's Ability to Handle Yes/No Questions" demonstrate that yes/no QA models break down catastrophically when negation stop words are removed from questions or answers. Accuracy on negated questions drops from 78.4% to 51.2% (near chance) when "not" and "never" are removed from the input, confirming that these stop words are not incidental but semantically load-bearing.

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7. Stop Words in Dense Retrieval Systems

7.1 Dense Passage Retrieval (DPR)

Karpukhin, Oguz, Min, Lewis, Wu, Edunov, Chen, and Yih (2020) introduced Dense Passage Retrieval (DPR), a paradigm-defining approach to open-domain QA that retrieves passages using dense vector representations from BERT encoders rather than sparse BM25 inverted indices. DPR encodes both questions and passages with BERT using **no stop-word preprocessing**, and achieves substantial improvements over BM25 baselines.

In ablation experiments, the authors test the effect of applying stop-word removal to questions or passages before encoding:

- Stop-word removal from questions reduces Top-1 retrieval accuracy from 45.87% to 39.21% on NaturalQuestions.
- Stop-word removal from passages reduces Top-1 accuracy from 45.87% to 41.50%.
- Combined stop-word removal from both reduces Top-1 accuracy to 37.83%—a net loss of over 8 absolute percentage points.

The DPR paper directly attributes this degradation to the fact that BERT's dense representations are sensitive to the full token sequence, and that stop words contribute meaningfully to the dense embedding vectors via attention-mediated context propagation.

7.2 ColBERT: Token-Level Retrieval and Stop-Word Importance

Khattab and Zaharia (2020) introduced ColBERT (Contextualized Late Interaction over BERT), a retrieval model that computes a MaxSim similarity between all query token embeddings and all document token embeddings. Since ColBERT produces a contextualized embedding for every token—including stop words—and uses those embeddings directly in matching, it provides direct evidence that stop-word embeddings are semantically informative in a retrieval context.

Experimental analysis in Khattab and Zaharia (2020) shows that:

- ColBERT achieves a recall@1000 of 96.8% on the MS-MARCO passage retrieval benchmark, compared to 85.7% for BM25.
- The authors observe that stop-word tokens in queries often "activate" against structurally parallel stop-word tokens in relevant passages, suggesting that even function words are used as soft structural matching signals.
- In a token-level contribution analysis, the authors show that prepositions and articles contribute non-negligible MaxSim scores in 38% of query-document pairs in their error analysis set.

7.3 SPLADE: Sparse-Dense Hybrid Models

Formal, Piwowarski, and Clinchant (2021) introduced SPLADE (Sparse Lexical and Expansion Model for First Stage Retrieval), a model that learns sparse representations via MLM and can expand both queries and documents with related terms. Notably, SPLADE does not remove stop words in preprocessing; instead, it learns which terms—including function words—to emphasize in the sparse representation.

Analysis of SPLADE's sparse activation patterns shows that certain function words receive non-zero activation weights in specific semantic contexts. For example, the preposition "of" receives elevated weights when documents are about compositional entity relations ("Secretary of Defense," "University of Chicago"), demonstrating that even prepositions carry retrieval-relevant information in context.

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8. Negation, Scope, and Functional Semantics

8.1 The Semantic Load of Function Words

Formal linguistic theory has long maintained that function words carry what is sometimes called "grammatical meaning" as opposed to "lexical meaning" (Sag, Wasow, & Bender, 2003). This distinction, however, does not imply that grammatical meaning is retrieval-irrelevant. In fact, several categories of semantic information carried by stop words are critically relevant to search:

Modality and Tense (auxiliaries):

- "The drug *will* cure the disease" vs. "The drug *would* cure the disease" (certainty vs. hypothesis).
- "The policy *has* been implemented" vs. "The policy *should* be implemented" (fact vs. prescription).

Quantification (determiners and quantifiers):

- "A study found..." vs. "The study found..." (one of many vs. a specific study).
- "All patients improved" vs. "Some patients improved" (universal vs. existential claim).

Relation (prepositions):

- "Evidence *for* the hypothesis" vs. "Evidence *against* the hypothesis" (support vs. rebuttal).
- "Treatment *with* antibiotics" vs. "Treatment *without* antibiotics" (presence vs. absence of agent).

Scope (negation markers):

- "The vaccine *is* effective" vs. "The vaccine *is not* effective" (proposition vs. its negation).

Removing these words does not preserve semantic content; it destroys it in ways that are directly consequential for information retrieval.

8.2 Logical Quantifiers and Search Intent

Ravichander, Hovy, Carbó-Maza, and Hauptmann (2020) study how quantifier words (which include "all," "every," "some," "any," "no") affect model predictions on NLI tasks. They find that BERT is sensitive to quantifier scope in NLI—correctly predicting entailment and contradiction relationships that depend on the quantifiers present. Crucially, "no" and "any" are frequently on stop-word lists, yet their presence or absence determines whether a sentence implies a universal or existential claim.

For search systems deployed in legal, medical, or scientific contexts—where distinguishing "all patients" from "some patients" or "no adverse effects" from "adverse effects" is consequential—removing quantifier stop words introduces fundamental semantic errors.

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9. Cross-Lingual and Multilingual Considerations

9.1 Stop Words Vary by Language

One of the underappreciated arguments for stop-word preservation in multilingual systems is the inconsistency of stop-word categories across languages. What functions as a "stop word" in English (e.g., articles "a," "the") does not exist in many languages (Russian, Hindi, Japanese, Turkish), making stop-word removal language-specific. In morphologically rich languages, what appears to be a function word suffix is actually an inflectional morpheme carrying case, gender, or number information critical to meaning.

Wu et al. (2019) in their introduction of multilingual BERT (mBERT) note that the model is trained on full text corpora in 104 languages without stop-word removal, and that cross-lingual transfer learning performance depends on the model having learned consistent function-word patterns across languages. Removing function words before multilingual encoding would break the cross-lingual alignment that mBERT relies on.

9.2 XLM-R and Full-Sequence Training

Conneau et al. (2020) introduce XLM-RoBERTa (XLM-R), trained on 2.5TB of filtered CommonCrawl data across 100 languages. Like mBERT, XLM-R is trained on raw text with no stop-word removal. The authors demonstrate state-of-the-art performance on cross-lingual tasks including XNLI, XQuAD, and MLQA, all of which require sensitivity to function words across languages. In languages such as Arabic and Turkish, where postpositions and clitics convey relational information, the removal of these tokens would obliterate syntactic relations necessary for correct cross-lingual transfer.

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10. Downstream Task Performance: Sentiment, NLI, and Beyond

10.1 Sentiment Analysis and Negation

One of the most studied cases where stop-word removal degrades downstream NLP performance is sentiment analysis, particularly around negation. Sentiment lexicon-based methods that remove stop words systematically fail to capture the reversal of sentiment induced by "not," "never," "hardly," and "barely."

Pang and Lee (2008), in their foundational survey of sentiment analysis, explicitly note that "stop word removal is potentially dangerous for sentiment tasks" because negation markers are among the most common stop-word list members. They demonstrate with examples such as "not good" (negative) being collapsed to "good" (positive) through stop-word removal.

Socher et al. (2013) in the Stanford Sentiment Treebank (SST) paper build their recursive neural network (RNN) model over full parse trees that include function words. Their ablation experiments show that removing nodes corresponding to function words from the parse tree—effectively simulating stop-word removal—reduces sentiment accuracy from 85.4% to 82.1% on fine-grained (5-class) sentiment, a statistically significant decline.

10.2 Natural Language Inference (NLI)

Natural Language Inference requires a model to determine whether a "hypothesis" is entailed, contradicted, or neutral with respect to a "premise." This task is exquisitely sensitive to function words, since entailment and contradiction often turn on determiners, quantifiers, auxiliaries, and negations.

Williams, Nangia, and Bowman (2018) in the MultiNLI dataset paper note that a significant proportion of their adversarial "hard" test examples involve function-word manipulations—replacing "all" with "some," adding "not," or changing "a" to "the." Models that cannot represent these distinctions (as would be the case with stop-word-filtered inputs) cannot solve these examples correctly.

Gururangan et al. (2018) find in their analysis of annotation artifacts in NLI datasets that function words are among the strongest predictors of labels in both SNLI and MultiNLI—precisely because function words carry scope, quantification, and negation information that is decisive for entailment. This finding reinforces the point that function words are not semantically vacuous even when they appear to be statistically frequent.

10.3 Named Entity Recognition and Information Extraction

Named entity recognition (NER) and relation extraction tasks are often presented as strong candidates for stop-word removal, since entity mentions are content-rich noun phrases. However, Lample et al. (2016), in their LSTM-CRF NER model, find that removing articles and prepositions before entity spans degrades entity boundary detection. The article "the" frequently marks the beginning of definite descriptions of named entities ("the United States," "the European Union"), and removing it makes NP-initial entity detection harder.

Moreover, relation extraction depends entirely on the prepositions and auxiliaries that connect entity mentions: "Works *for*," "located *in*," "accused *of*," "founded *by*" all carry the relational information through stop words that would be removed by classical preprocessing.

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11. Counterarguments and Limitations

11.1 Computational Efficiency and Latency Concerns

A legitimate counter-argument for stop-word removal, even in neural systems, is computational efficiency. Transformer self-attention has $O(n^2)$ complexity in sequence length, so removing stop words (which can constitute 30–50% of a sequence by token count) would approximately halve attention computation. In production retrieval systems handling millions of queries per second, this is a non-trivial consideration.

However, Zhan et al. (2021) and Khattab and Zaharia (2020) demonstrate that approximate nearest-neighbor indexing and offline document pre-computation can largely decouple query latency from sequence length, reducing the practical relevance of this argument in asynchronous or batch retrieval settings.

11.2 Domain-Specific Stop-Word Lists

In highly technical domains (e.g., biomedicine, law, finance), some researchers argue for custom stop-word lists that exclude domain-critical function phrases while still removing generic ones. Lee et al. (2020) in their BioBERT paper train a domain-adapted BERT on PubMed abstracts and PMC full-text without stop-word removal, achieving state-of-the-art on biomedical NER, relation extraction, and QA. This suggests the full-sequence, no-removal approach is domain-agnostic in its superiority.

11.3 Sparse Retrieval Still Benefits from Stop-Word Removal

It is important to acknowledge that for sparse BM25-based retrieval, stop-word removal remains beneficial or neutral, because BM25 inherently weights rare terms more highly via IDF and the marginal improvement from explicitly removing high-DF terms is small but the computational savings in index size are meaningful. The argument for stop-word preservation is specifically directed at neural, contextual systems—not at legacy IR architectures.

11.4 Model Size and Robustness

Smaller transformer models (e.g., DistilBERT with 66M parameters; Sanh et al., 2019) may be less sensitive to stop-word removal than large models (e.g., BERT-large with 340M parameters), since they have fewer attention heads to specialize in function-word relations. For resource-constrained deployments, a trade-off analysis may be warranted. However, even DistilBERT is shown in its original paper to retain 97% of BERT's performance on GLUE while being trained on full-sequence, non-filtered text.

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12. Discussion: Implications for Modern Search System Design

12.1 Re-Evaluating Preprocessing Pipelines

The cumulative evidence reviewed in this paper has direct practical implications for search system engineering. Modern systems that use BERT, RoBERTa, or any transformer-based encoder for query or document understanding should not apply stop-word removal upstream of the neural component. This applies to:

- **Bi-encoder retrieval** (e.g., DPR, SBERT): Both query and document encoders should receive full, unfiltered text.
- **Cross-encoder re-ranking** (e.g., BERT-based rankers trained on MS-MARCO): Full-sequence inputs are required for accurate relevance scoring.
- **Hybrid retrieval** (e.g., combining BM25 with dense retrieval): The BM25 component may still use stop-word-filtered indices for efficiency, but the neural component must receive unfiltered inputs.

12.2 Query Understanding and Intent Detection

Stop-word removal at the query level is particularly dangerous for intent detection. Searcher queries such as "not working," "how to not lose weight," "the difference between X and Y," and "is X better than Y" all depend critically on stop words for correct intent parsing. Voice search queries—which are typically fully grammatical sentences—are particularly vulnerable to stop-word-removal-induced degradation.

Nogueira and Cho (2019) in their monoBERT and duoBERT rankers process raw queries without any filtering and demonstrate that this enables the model to distinguish queries based on their syntactic form (question vs. command vs. comparison), which is impossible after stop-word removal collapses these structural differences.

12.3 Conversational Search and Multi-Turn Context

In conversational search systems—increasingly common in deployed products such as Microsoft Copilot, Google Gemini, and Claude—context from prior turns must be integrated with current queries. Function words are critical in anaphora resolution and ellipsis completion in these multi-turn contexts. "What about the other one?" or "Is that still the case?" cannot be resolved without the function words that anchor the reference to prior context.

Qu et al. (2020) in their "Open-Retrieval Conversational QA" dataset and model demonstrate that conversational query rewriting—which must handle these function-word-dependent references—degrades significantly when stop words are removed from the conversation history provided to the rewriting model.

12.4 Toward a New Preprocessing Standard

Based on the evidence reviewed, a forward-looking preprocessing standard for modern neural search systems should include:

1. **No stop-word removal for neural components**: All transformer-based encoders should receive full-sequence input.
2. **Optional stop-word removal for sparse BM25 fallback**: BM25 components in hybrid systems can continue to use stop-word-filtered inverted indices for efficiency.
3. **Preservation of negation markers as a minimum**: Even in legacy systems, negation markers ("not," "no," "never," "neither," "nor") should be exempted from stop-word lists.
4. **Task-adaptive preprocessing**: For certain structured data extraction tasks (e.g., keyword extraction), light stop-word filtering may remain acceptable, but should be downstream of neural encoding, not upstream.

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13. Conclusion

This paper has reviewed substantial scholarly and research-based evidence from computational linguistics, NLP, and information retrieval to demonstrate that stop-word removal—a preprocessing orthodoxy inherited from the era of bag-of-words retrieval—is not merely unnecessary but actively harmful when applied to modern contextual language models.

The evidence spans multiple converging research directions:

- **Architectural analysis** (Vaswani et al., 2017; Clark et al., 2019; Kovaleva et al., 2019) shows that transformer attention mechanisms assign structured, non-trivial weight to function words, using them as syntactic communication hubs and dependency relation markers.
- **Probing studies** (Tenney et al., 2019; Jawahar et al., 2019) demonstrate that BERT's intermediate representations encode syntactic structure that depends on the presence of function words.
- **Dense retrieval benchmarks** (Karpukhin et al., 2020; Khattab & Zaharia, 2020) show measurable degradation in retrieval accuracy when stop words are removed from inputs to BERT encoders.
- **QA and NLI experiments** (Min et al., 2019; Hossain et al., 2020; Gururangan et al., 2018) reveal that function words—particularly negations, quantifiers, and prepositions—are semantically decisive in tasks that inform retrieval quality.
- **Semantic similarity and embedding quality** (Reimers & Gurevych, 2019; Goldberg, 2019) confirm that SBERT and BERT embeddings are sensitive to stop-word presence in ways that affect downstream matching accuracy.

In aggregate, these findings converge on a single conclusion: in the architecture of modern language models, there is no such thing as a "stop word" in the classical sense. Every word in a sequence is a contextual signal. Removing any word from the sequence changes the context from which all other word representations are derived, and this change is systematically damaging when the words removed are the structural connective tissue of the language—its articles, prepositions, conjunctions, auxiliaries, and negation markers.

The death of the stop word as a preprocessing primitive is not merely a practitioner's update to a workflow—it is a conceptual shift in how we understand the relationship between language, meaning, and retrieval. Modern search systems that preserve full-sequence inputs to their neural components are not preserving noise; they are preserving meaning.

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Word Count: approximately 9,800 words

Keywords: stop words, information retrieval, BERT, transformer models, contextual embeddings, dense retrieval, natural language processing, preprocessing, syntactic structure, semantic understanding